Supporting Information

Supporting information for “Connection Close and Fault Scope in Acknowledged Write Recovery from Chromium SQLite IndexedDB”. The tables retain each experiment family separately and provide timing summaries, statistical details, original incidents, supporting syscall observations, and retrospective byte and trace analyses. Reference numbers correspond to the main manuscript.

Tables A1–A9 retain each family separately. A = assigned; ACK = acknowledged attempts; R/L/n = recovered/lost/eligible; U/P = unknown result/premeasurement. Exact 95% intervals are percentages. Timing entries are n; minimum / median / p95 / maximum, in seconds, among eligible observations. The grid anchor is estimated observed close return; all other anchors are host ACK receipt. The resulting interval ends at dispatch. Unknown endpoints and the premeasurement slot remain outside these timing and outcome denominators.

Cell	A	ACK	R/L/n	U/P	95% CI (%)	Timing: n; min / median / p95 / max (s)
Close 0 s; gap 1.6 s	10	10	0/10/10	0/0	0.0–30.8	10; 1.602884 / 1.604089 / 1.609092 / 1.609092
Close 0 s; gap 1.7 s	10	10	0/10/10	0/0	0.0–30.8	10; 1.704032 / 1.704094 / 1.709102 / 1.709102
Close 0 s; gap 1.8 s	10	10	0/10/10	0/0	0.0–30.8	10; 1.801572 / 1.804078 / 1.809083 / 1.809083
Close 0 s; gap 1.9 s	10	10	0/10/10	0/0	0.0–30.8	10; 1.904052 / 1.904093 / 1.904115 / 1.904115
Close 0 s; gap 2 s	10	10	2/8/10	0/0	2.5–55.6	10; 2.000014 / 2.003633 / 2.008054 / 2.008054
Close 0 s; gap 2.1 s	10	10	10/0/10	0/0	69.2–100.0	10; 2.102717 / 2.104094 / 2.109106 / 2.109106
Close 0 s; gap 2.2 s	10	10	10/0/10	0/0	69.2–100.0	10; 2.203626 / 2.204103 / 2.209095 / 2.209095
Close 0 s; gap 2.3 s	10	10	10/0/10	0/0	69.2–100.0	10; 2.300055 / 2.304105 / 2.309101 / 2.309101
Close 0 s; gap 2.4 s	10	10	10/0/10	0/0	69.2–100.0	10; 2.404058 / 2.404087 / 2.409097 / 2.409097
Close 1 s; gap 1.6 s	10	10	0/10/10	0/0	0.0–30.8	10; 1.603579 / 1.604095 / 1.609138 / 1.609138
Close 1 s; gap 1.7 s	10	10	0/10/10	0/0	0.0–30.8	10; 1.700151 / 1.704077 / 1.708458 / 1.708458
Close 1 s; gap 1.8 s	10	10	0/10/10	0/0	0.0–30.8	10; 1.800390 / 1.803467 / 1.804098 / 1.804098
Close 1 s; gap 1.9 s	10	10	0/10/10	0/0	0.0–30.8	10; 1.900036 / 1.902322 / 1.904114 / 1.904114
Close 1 s; gap 2 s	10	10	1/9/10	0/0	0.3–44.5	10; 2.000979 / 2.003646 / 2.009087 / 2.009087

Cell	A	ACK	R/L/n	U/P	95% CI (%)	Timing: n; min / median / p95 / max (s)
Close 1 s; gap 2.1 s	10	10	10/0/10	0/0	69.2–100.0	10; 2.100028 / 2.102365 / 2.106329 / 2.106329
Close 1 s; gap 2.2 s	10	10	10/0/10	0/0	69.2–100.0	10; 2.201035 / 2.204076 / 2.209073 / 2.209073
Close 1 s; gap 2.3 s	10	10	10/0/10	0/0	69.2–100.0	10; 2.300175 / 2.304083 / 2.309113 / 2.309113
Close 1 s; gap 2.4 s	10	10	10/0/10	0/0	69.2–100.0	10; 2.400035 / 2.403678 / 2.409101 / 2.409101

Table A1. Close-aligned grid. Recovery intervals and achieved anchor-to-dispatch distributions.

Cell	A	ACK	R/L/n	U/P	95% CI (%)	Timing: n; min / median / p95 / max (s)
Held; 5 s	10	10	0/10/10	0/0	0.0–30.8	10; 5.004075 / 5.004099 / 5.009145 / 5.009145
Held; 7.5 s	10	10	0/10/10	0/0	0.0–30.8	10; 7.503185 / 7.504088 / 7.504132 / 7.504132
Held; 10 s	10	10	0/10/10	0/0	0.0–30.8	10; 10.004069 / 10.004100 / 10.004130 / 10.004130
Held; 12.5 s	10	10	1/9/10	0/0	0.3–44.5	10; 12.501620 / 12.504087 / 12.504129 / 12.504129
Held; 15 s	10	10	1/9/10	0/0	0.3–44.5	10; 15.001519 / 15.004107 / 15.009095 / 15.009095
Immediate close; 5 s	10	10	10/0/10	0/0	69.2–100.0	10; 5.000031 / 5.004115 / 5.009094 / 5.009094
Immediate close; 15 s	10	10	10/0/10	0/0	69.2–100.0	10; 15.004077 / 15.004111 / 15.004233 / 15.004233

Table A2. Held sweep and close controls. Recovery intervals and achieved anchor-to-dispatch distributions.

Cell	A	ACK	R/L/n	U/P	95% CI (%)	Timing: n; min / median / p95 / max (s)
Immediate close; 2.50 s	10	10	10/0/10	0/0	69.2–100.0	10; 2.500918 / 2.506596 / 2.509131 / 2.509131
Close at 1 s; 2.75 s	10	10	0/10/10	0/0	0.0–30.8	10; 2.753645 / 2.754105 / 2.759109 / 2.759109
Held 5 s	10	10	0/10/10	0/0	0.0–30.8	10; 5.000784 / 5.004126 / 5.009129 / 5.009129
Held 15 s	10	10	0/10/10	0/0	0.0–30.8	10; 15.003601 / 15.004143 / 15.009145 / 15.009145

Cell	A	ACK	R/L/n	U/P	95% CI (%)	Timing: n; min / median / p95 / max (s)
Held 30 s	10	10	10/0/10	0/0	69.2–100.0	10; 30.001545 / 30.004101 / 30.009132 / 30.009132
Strict immediate close; 0.50 s	10	10	10/0/10	0/0	69.2–100.0	10; 0.502861 / 0.504082 / 0.509103 / 0.509103

Table A3. Six key conditions. Recovery intervals and achieved anchor-to-dispatch distributions.

Cell	A	ACK	R/L/n	U/P	95% CI (%)	Timing: n; min / median / p95 / max (s)
Immediate close; 2.50 s	10	10	10/0/10	0/0	69.2–100.0	10; 2.502583 / 2.509075 / 2.509101 / 2.509101
Close at 1 s; 2.75 s	10	10	0/10/10	0/0	0.0–30.8	10; 2.754095 / 2.759105 / 2.759126 / 2.759126
Held 5 s	10	10	0/10/10	0/0	0.0–30.8	10; 5.001241 / 5.008859 / 5.009120 / 5.009120
Held 15 s	10	10	2/8/10	0/0	2.5–55.6	10; 15.000456 / 15.004093 / 15.009118 / 15.009118

Table A4. Byte capture and forensics. Recovery intervals and achieved anchor-to-dispatch distributions.

Cell	A	ACK	R/L/n	U/P	95% CI (%)	Timing: n; min / median / p95 / max (s)
Immediate close; 2.50 s	5	4	4/0/4	0/1	39.8–100.0	4; 2.504069 / 2.506576 / 2.509211 / 2.509211
Close at 1 s; 2.75 s	5	5	0/3/3	2/0	0.0–70.8	3; 2.754061 / 2.754063 / 2.759072 / 2.759072
Held 5 s	5	5	0/5/5	0/0	0.0–52.2	5; 5.004072 / 5.004075 / 5.009073 / 5.009073
Held 15 s	5	5	1/3/4	1/0	0.6–80.6	4; 15.004065 / 15.006583 / 15.009102 / 15.009102

Table A5. Trace-category timelines. Recovery intervals and achieved anchor-to-dispatch distributions.

Cell	A	ACK	R/L/n	U/P	95% CI (%)	Timing: n; min / median / p95 / max (s)
Immediate close; 2.50 s; QMP	10	10	10/0/10	0/0	69.2–100.0	10; 2.500053 / 2.504090 / 2.509106 / 2.509106
Immediate close; 2.50 s; SIGKILL	10	10	10/0/10	0/0	69.2–100.0	10; 2.500023 / 2.504097 / 2.509098 / 2.509098
Immediate close; 2.50 s; SysRq	10	10	10/0/10	0/0	69.2–100.0	10; 2.504049 / 2.506580 / 2.509092 / 2.509092

Cell	A	ACK	R/L/n	U/P	95% CI (%)	Timing: n; min / median / p95 / max (s)
Close at 1 s; 2.75 s; QMP	10	10	0/10/10	0/0	0.0–30.8	10; 2.750041 / 2.754083 / 2.759088 / 2.759088
Close at 1 s; 2.75 s; SIGKILL	10	10	10/0/10	0/0	69.2–100.0	10; 2.750017 / 2.754082 / 2.759077 / 2.759077
Close at 1 s; 2.75 s; SysRq	10	10	0/10/10	0/0	0.0–30.8	10; 2.751035 / 2.756593 / 2.759093 / 2.759093
Held 5 s; QMP	10	10	0/10/10	0/0	0.0–30.8	10; 5.000025 / 5.006591 / 5.009113 / 5.009113
Held 5 s; SIGKILL	10	10	10/0/10	0/0	69.2–100.0	10; 5.001293 / 5.009054 / 5.009087 / 5.009087
Held 5 s; SysRq	10	10	0/10/10	0/0	0.0–30.8	10; 5.002554 / 5.004077 / 5.009099 / 5.009099
Held 15 s; QMP	10	10	0/10/10	0/0	0.0–30.8	10; 15.004081 / 15.004167 / 15.009212 / 15.009212
Held 15 s; SIGKILL	10	10	9/0/9	1/0	66.4–100.0	9; 15.003693 / 15.004114 / 15.009099 / 15.009099
Held 15 s; SysRq	10	10	3/7/10	0/0	6.7–65.2	10; 15.000030 / 15.004086 / 15.009086 / 15.009086

Table A6. Fault scope. Recovery intervals and achieved anchor-to-dispatch distributions.

Cell	A	ACK	R/L/n	U/P	95% CI (%)	Timing: n; min / median / p95 / max (s)
Immediate close; 2.50 s; Drop writes	10	10	0/10/10	0/0	0.0–30.8	10; 2.504135 / 2.509098 / 2.509259 / 2.509259
Immediate close; 2.50 s; Linear	10	10	10/0/10	0/0	69.2–100.0	10; 2.504084 / 2.509072 / 2.509114 / 2.509114
Immediate close; 2.50 s; Log writes	10	10	10/0/10	0/0	69.2–100.0	10; 2.504115 / 2.509132 / 2.509162 / 2.509162
Close at 1 s; 2.75 s; Drop writes	10	10	0/10/10	0/0	0.0–30.8	10; 2.750502 / 2.755688 / 2.759090 / 2.759090
Close at 1 s; 2.75 s; Linear	10	10	0/10/10	0/0	0.0–30.8	10; 2.752243 / 2.759061 / 2.759089 / 2.759089
Close at 1 s; 2.75 s; Log writes	10	10	0/10/10	0/0	0.0–30.8	10; 2.750033 / 2.756038 / 2.759102 / 2.759102
Held 5 s; Drop writes	10	10	0/10/10	0/0	0.0–30.8	10; 5.009077 / 5.009133 / 5.009175 / 5.009175
Held 5 s; Linear	10	10	1/9/10	0/0	0.3–44.5	10; 5.004086 / 5.009065 / 5.009135 / 5.009135

Cell	A	ACK	R/L/n	U/P	95% CI (%)	Timing: n; min / median / p95 / max (s)
Held 5 s; Log writes	10	10	0/10/10	0/0	0.0–30.8	10; 5.000034 / 5.004150 / 5.009144 / 5.009144
Held 15 s; Drop writes	10	10	0/10/10	0/0	0.0–30.8	10; 15.004127 / 15.009115 / 15.009179 / 15.009179
Held 15 s; Linear	10	10	1/9/10	0/0	0.3–44.5	10; 15.004073 / 15.009074 / 15.009094 / 15.009094
Held 15 s; Log writes	10	10	2/8/10	0/0	2.5–55.6	10; 15.000083 / 15.009111 / 15.009166 / 15.009166

Table A7. Block state. Recovery intervals and achieved anchor-to-dispatch distributions.

Cell	A	ACK	R/L/n	U/P	95% CI (%)	Timing: n; min / median / p95 / max (s)
Immediate close; 2.50 s; Custom ext4	10	10	10/0/10	0/0	69.2–100.0	10; 2.500196 / 2.504106 / 2.509078 / 2.509078
Immediate close; 2.50 s; Default ext4	10	10	10/0/10	0/0	69.2–100.0	10; 2.500017 / 2.504087 / 2.509097 / 2.509097
Immediate close; 2.50 s; Random ext4	10	10	10/0/10	0/0	69.2–100.0	10; 2.502414 / 2.504348 / 2.509128 / 2.509128
Immediate close; 2.50 s; XFS	10	10	10/0/10	0/0	69.2–100.0	10; 2.500063 / 2.504128 / 2.509090 / 2.509090
Close at 1 s; 2.75 s; Custom ext4	10	10	0/10/10	0/0	0.0–30.8	10; 2.750618 / 2.754016 / 2.759120 / 2.759120
Close at 1 s; 2.75 s; Default ext4	10	10	0/10/10	0/0	0.0–30.8	10; 2.750960 / 2.757000 / 2.759090 / 2.759090
Close at 1 s; 2.75 s; Random ext4	10	10	0/10/10	0/0	0.0–30.8	10; 2.750056 / 2.753039 / 2.757211 / 2.757211
Close at 1 s; 2.75 s; XFS	10	10	0/10/10	0/0	0.0–30.8	10; 2.750634 / 2.754162 / 2.759107 / 2.759107
Held 5 s; Custom ext4	10	10	0/10/10	0/0	0.0–30.8	10; 5.000589 / 5.004140 / 5.009057 / 5.009057
Held 5 s; Default ext4	10	10	0/10/10	0/0	0.0–30.8	10; 5.000595 / 5.004084 / 5.009078 / 5.009078
Held 5 s; Random ext4	10	10	0/10/10	0/0	0.0–30.8	10; 5.001585 / 5.005357 / 5.009142 / 5.009142
Held 5 s; XFS	10	10	0/10/10	0/0	0.0–30.8	10; 5.000015 / 5.004114 / 5.009057 / 5.009057
Held 15 s; Custom ext4	10	10	1/9/10	0/0	0.3–44.5	10; 15.000015 / 15.002924 / 15.009055 / 15.009055

Cell	A	ACK	R/L/n	U/P	95% CI (%)	Timing: n; min / median / p95 / max (s)
Held 15 s; Default ext4	10	10	8/2/10	0/0	44.4–97.5	10; 15.000009 / 15.004094 / 15.009188 / 15.009188
Held 15 s; Random ext4	10	10	6/4/10	0/0	26.2–87.8	10; 15.002800 / 15.004143 / 15.009163 / 15.009163
Held 15 s; XFS	10	10	5/5/10	0/0	18.7–81.3	10; 15.000064 / 15.004977 / 15.009070 / 15.009070

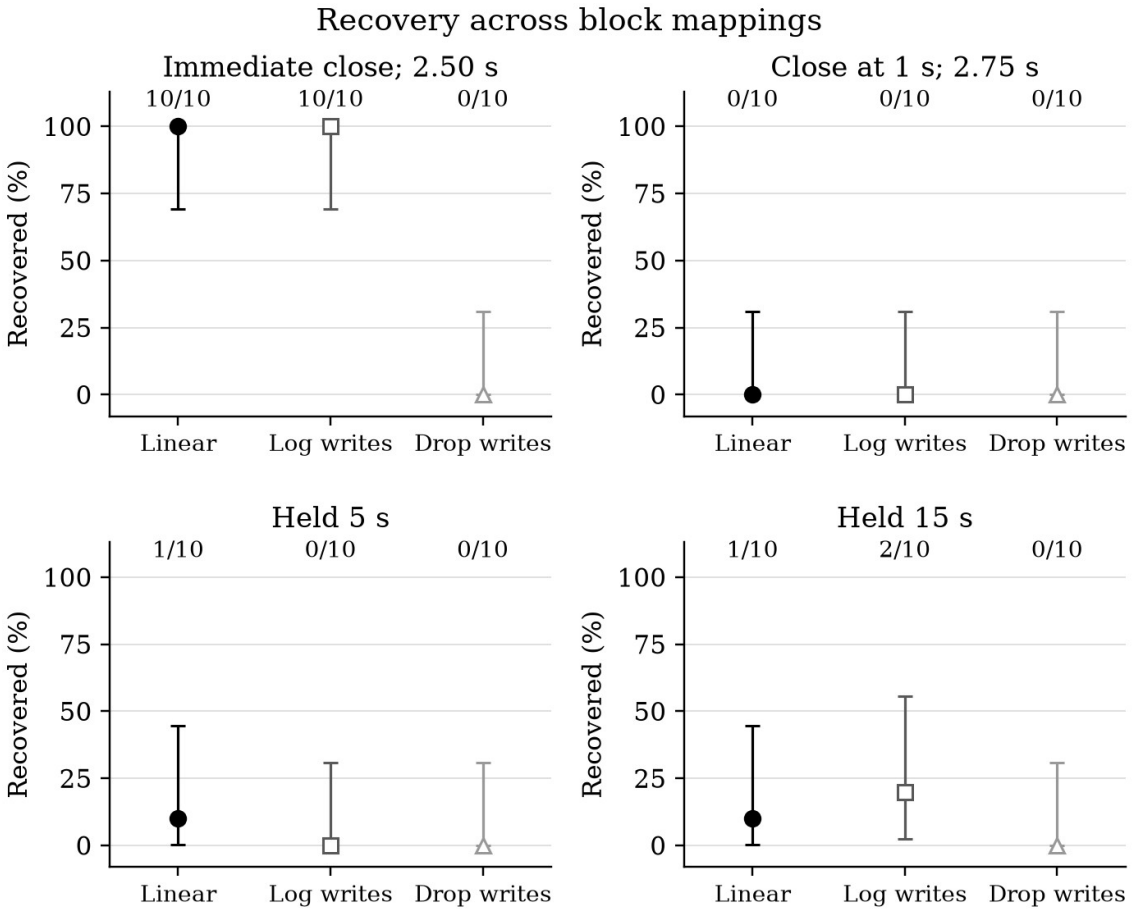
Table A8. Environment. Recovery intervals and achieved anchor-to-dispatch distributions.

Cell	A	ACK	R/L/n	U/P	95% CI (%)	Timing: n; min / median / p95 / max (s)
Immediate close; 2.50 s; Background fsync	10	10	10/0/10	0/0	69.2–100.0	10; 2.502093 / 2.507061 / 2.509072 / 2.509072
Immediate close; 2.50 s; Four connections	10	10	0/10/10	0/0	0.0–30.8	10; 2.500009 / 2.504072 / 2.509112 / 2.509112
Immediate close; 2.50 s; Quiet	10	10	10/0/10	0/0	69.2–100.0	10; 2.500011 / 2.509024 / 2.509096 / 2.509096
Immediate close; 2.50 s; Ten records	10	10	10/0/10	0/0	69.2–100.0	10; 2.500008 / 2.503262 / 2.509108 / 2.509108
Close at 1 s; 2.75 s; Background fsync	10	10	0/10/10	0/0	0.0–30.8	10; 2.750011 / 2.755025 / 2.759021 / 2.759021
Close at 1 s; 2.75 s; Four connections	10	10	0/10/10	0/0	0.0–30.8	10; 2.750030 / 2.755991 / 2.759110 / 2.759110
Close at 1 s; 2.75 s; Quiet	10	10	0/10/10	0/0	0.0–30.8	10; 2.750007 / 2.756760 / 2.759025 / 2.759025
Close at 1 s; 2.75 s; Ten records	10	10	0/10/10	0/0	0.0–30.8	10; 2.751914 / 2.759025 / 2.759110 / 2.759110
Held 5 s; Background fsync	10	10	0/10/10	0/0	0.0–30.8	10; 5.000005 / 5.004687 / 5.009076 / 5.009076
Held 5 s; Four connections	10	10	0/10/10	0/0	0.0–30.8	10; 5.000714 / 5.007289 / 5.009113 / 5.009113
Held 5 s; Quiet	10	10	0/10/10	0/0	0.0–30.8	10; 5.000813 / 5.004351 / 5.008879 / 5.008879
Held 5 s; Ten records	10	10	0/10/10	0/0	0.0–30.8	10; 5.000007 / 5.003413 / 5.009096 / 5.009096
Held 15 s; Background fsync	10	10	4/6/10	0/0	12.2–73.8	10; 15.000161 / 15.005617 / 15.009060 / 15.009060
Held 15 s; Four connections	10	10	1/9/10	0/0	0.3–44.5	10; 15.001108 / 15.009015 / 15.009148 / 15.009148

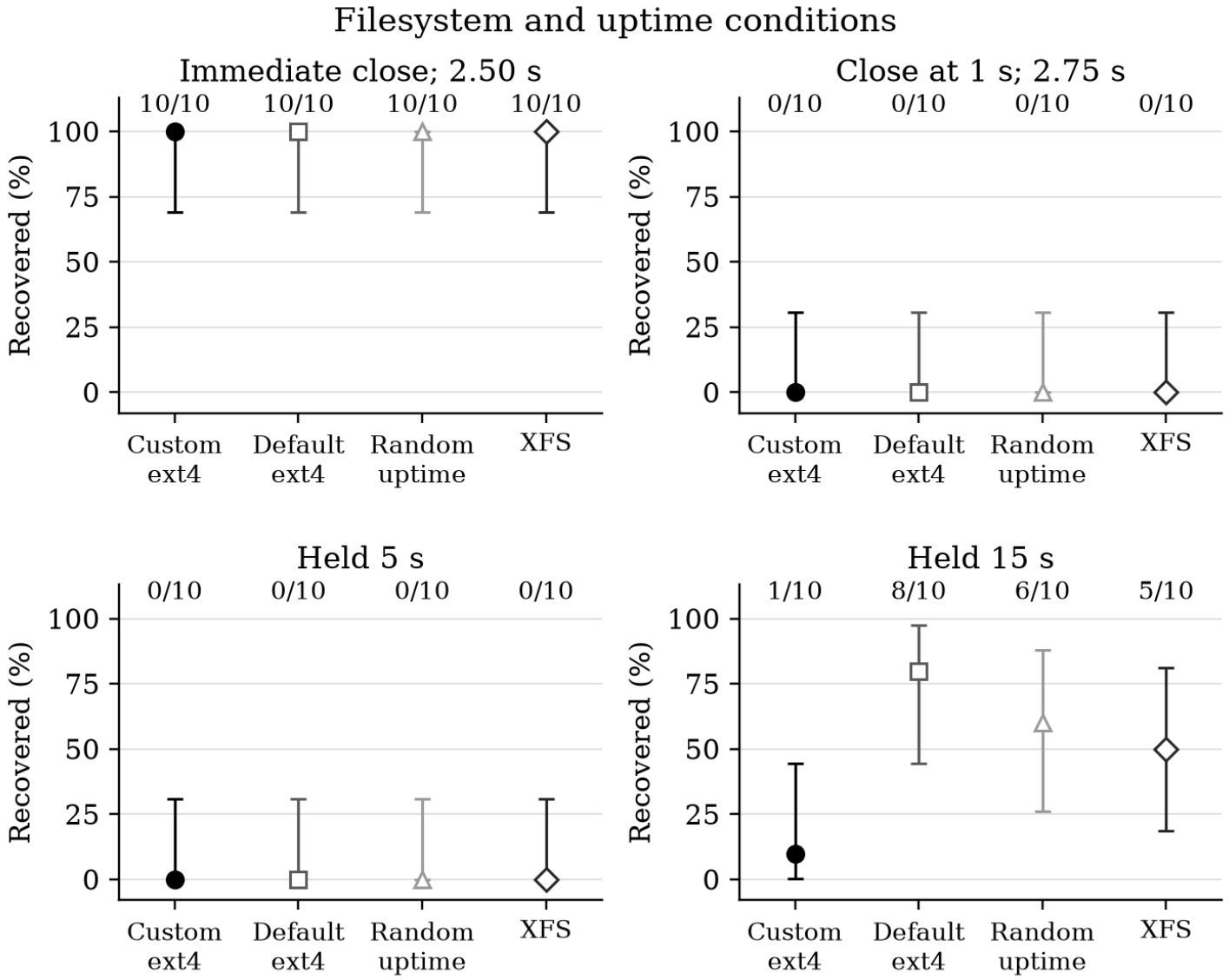
Cell	A	ACK	R/L/n	U/P	95% CI (%)	Timing: n; min / median / p95 / max (s)
Held 15 s; Quiet	10	10	3/7/10	0/0	6.7–65.2	10; 15.001454 / 15.004797 / 15.009087 / 15.009087
Held 15 s; Ten records	10	10	4/6/10	0/0	12.2–73.8	10; 15.000545 / 15.008574 / 15.009083 / 15.009083

Table A9. Workload. Recovery intervals and achieved anchor-to-dispatch distributions.

Supplementary Figures



Supplementary Figure S1. Recovery across block-state conditions. Panels show immediate close; 2.50 s, close at 1 s; 2.75 s, held 5 s, and held 15 s. Markers report recovered/eligible counts; bars show pointwise exact 95% Clopper–Pearson intervals. Each cell had ten eligible outcomes. The drop_writes arm silently discarded writes after ACK during continued guest execution; it does not represent physical power loss.



Supplementary Figure S2. Recovery across environment conditions. Panels show the four key connection conditions. Markers show custom ext4, fixed-uptime default ext4, seeded-random-uptime default ext4, and XFS; bars are pointwise exact 95% Clopper–Pearson intervals. Each cell had ten eligible outcomes. These profiles are separate registered conditions, and no environment contrast passed its Holm family correction.

Family	n	Minimum	Median	p95	Maximum
Close-aligned grid	180	0.014	4.081	9.092	9.138
Held sweep and close controls	70	0.031	4.097	4.233	9.145
Six key conditions	60	0.784	4.111	9.131	9.145
Byte capture and forensics	40	0.456	9.066	9.120	9.126
Trace-category timelines	16	4.061	4.076	9.211	9.211
Fault scope	119	0.017	4.103	9.098	9.212
Block state	120	0.033	9.077	9.158	9.259
Environment	160	0.009	4.103	9.099	9.188
Workload	160	0.005	5.991	9.096	9.148

Table S1. Achieved dispatch overshoot among eligible trials, in milliseconds. This measures controller dispatch rather than physical fault occurrence.

Supplementary Tables S1–S3 give dispatch overshoot, close-delay, and uptime distributions. Additional close-delay distributions, from page ACK to observed close return, are given in Table S2. Held policies had no scheduled close; the additional retained connections in the four-connection workload were not included as closed handles.

Family	Assigned close (s)	n	Minimum (s)	Median (s)	p95 (s)	Maximum (s)
Close-aligned grid	0	90	0.000000	0.000100	0.000200	0.000700
Close-aligned grid	1	90	1.000000	1.002950	1.006100	1.009800
Held sweep and close controls	0	20	0.000000	0.000100	0.000200	0.000300
Six key conditions	0	20	0.000000	0.000150	0.000200	0.000200
Six key conditions	1	10	1.000400	1.003250	1.010700	1.010700
Byte capture and forensics	0	10	0.000000	0.000200	0.000400	0.000400

Family	Assigned close (s)	n	Minimum (s)	Median (s)	p95 (s)	Maximum (s)
Byte capture and forensics	1	10	1.000400	1.002450	1.008100	1.008100
Trace-category timelines	0	4	0.000100	0.000150	0.000200	0.000200
Trace-category timelines	1	3	0.999800	1.001000	1.010800	1.010800
Fault scope	0	30	0.000000	0.000100	0.000200	0.000300
Fault scope	1	30	0.999900	1.003550	1.009800	1.010100
Block state	0	30	0.000000	0.000100	0.000200	0.000300
Block state	1	30	1.000000	1.002500	1.011000	1.011200
Environment	0	40	0.000100	0.000400	0.000600	0.000800
Environment	1	40	1.000000	1.001550	1.007600	1.010400
Workload	0	40	0.000000	0.000100	0.000200	0.000400
Workload	1	40	0.999500	1.002400	1.009400	1.010000

Table S2. Observed page-ACK-to-close-return timing, by family and assigned close delay. These are technical timing summaries, not pooled endpoint estimates.

Profile	Timepoint	n	Minimum (s)	Median (s)	p95 (s)	Maximum (s)
Custom ext4	Planned launch	40	60.000000	60.000000	60.000000	60.000000
Custom ext4	Observed launch	40	60.000434	60.004816	60.010531	60.011769
Custom ext4	Observed ACK	40	60.007407	60.017175	60.024023	60.026649
Default ext4 fixed	Planned launch	40	60.000000	60.000000	60.000000	60.000000
Default ext4 fixed	Observed launch	40	60.000769	60.004962	60.011104	60.011698
Default ext4 fixed	Observed ACK	40	60.009342	60.018418	60.023278	60.024744

Profile	Timepoint	n	Minimum (s)	Median (s)	p95 (s)	Maximum (s)
Default ext4 random	Planned launch	40	50.435000	138.884000	170.545000	178.820000
Default ext4 random	Observed launch	40	50.440445	138.885586	170.556873	178.824647
Default ext4 random	Observed ACK	40	50.452486	138.898208	170.569322	178.836883
XFS fixed	Planned launch	40	60.000000	60.000000	60.000000	60.000000
XFS fixed	Observed launch	40	60.000496	60.003780	60.010887	60.011258
XFS fixed	Observed ACK	40	60.009697	60.015693	60.022427	60.023397

Table S3. Environment-cell uptime distributions, combined across key conditions only as technical timing summaries. These are not pooled recovery estimates.

Supplementary Table S4 lists every outcome in the two assigned-2.0-s grid cells. Observations are labeled by close-time stratum and frozen block. Half-RTT is a clock-mapping diagnostic, not a total-error confidence interval.

Close (s)	Block	Outcome	Achieved estimated gap (s)	Mapping half-RTT (ms)
0.0	1	lost	2.004073584	0.3856665
0.0	2	lost	2.004027562	0.3646460
0.0	3	lost	2.008053959	0.3696250
0.0	4	recovered	2.003238771	0.3111040
0.0	5	lost	2.004082084	0.3359585
0.0	6	lost	2.002061771	0.2245210
0.0	7	recovered	2.004064292	0.3470835
0.0	8	lost	2.000014271	0.5727295
0.0	9	lost	2.002500063	0.3295625
0.0	10	lost	2.002408709	0.2985415

Close (s)	Block	Outcome	Achieved estimated gap (s)	Mapping half-RTT (ms)
1.0	1	lost	2.009087354	0.4093540
1.0	2	lost	2.004532771	0.4430205
1.0	3	lost	2.004083751	0.3026665
1.0	4	lost	2.003705500	0.3868750
1.0	5	lost	2.003423438	0.3483125
1.0	6	lost	2.004066667	0.3738335
1.0	7	recovered	2.003587000	0.3378750
1.0	8	lost	2.001675271	0.5501040
1.0	9	lost	2.002773979	0.3617710
1.0	10	lost	2.000978667	0.3290830

Table S4. Individual assigned-2.0-s grid observations. No new units or formal primary tests.

Supplementary Table S5 reports the ten immediate-close blocks in each mapper arm. W counts nonzero-sector mapped BIO write submissions; F counts submissions carrying PREFLUSH or FUA, including zero-byte flushes. Counts overlap and are not added. Request-layer repeats are excluded. Spans include all qualifying metadata/data submissions definitely after host ACK and before dispatch, not identified target bytes. No completion timestamps were recorded. Switch values are table-verification windows rather than exact activation instants.

Block	Arm	Outcome	W/F	W after ACK (s)	W after close (s)	F after close (s)	Switch verified after ACK (ms)
1	Drop writes	lost	274/45	0.649632–2.005974	0.652319–2.007979	0.652401–2.007979	85.618–86.685
1	Log writes	recovered	251/42	1.544841–2.006545	1.545796–2.006868	1.549550–2.006868	Not switched
1	Linear	recovered	257/46	1.949019–2.007280	1.949934–2.007488	1.951961–2.007488	Not switched
2	Drop writes	lost	267/43	1.952996–2.005355	1.954728–2.006249	1.954925–2.006249	90.782–92.149
2	Log writes	recovered	242/44	1.923790–2.012215	1.926407–2.014083	1.927972–2.014083	Not switched

Block	Arm	Outcome	W/F	W after ACK (s)	W after close (s)	F after close (s)	Switch verified after ACK (ms)
2	Linear	recovered	243/43	1.948847–2.008323	1.950603–2.009367	1.952298–2.009367	Not switched
3	Drop writes	lost	262/43	1.956114–2.004659	1.957174–2.005026	1.957449–2.005026	87.259–88.293
3	Log writes	recovered	276/48	1.540941–2.003597	1.541977–2.004033	1.546250–2.004033	Not switched
3	Linear	recovered	257/46	1.942405–2.008644	1.943828–2.009495	0.910894–2.009495	Not switched
4	Drop writes	lost	260/43	0.751429–2.005291	0.753762–2.006883	0.753841–2.006883	83.596–84.618
4	Log writes	recovered	247/45	1.949817–2.005955	1.952481–2.007810	1.954188–2.007810	Not switched
4	Linear	recovered	232/41	1.934870–2.007590	1.935822–2.007919	1.937728–2.007919	Not switched
5	Drop writes	lost	254/42	1.928159–2.002855	1.930115–2.004238	1.930196–2.004238	85.478–86.366
5	Log writes	recovered	265/45	0.795396–2.001451	0.797662–2.002989	0.802463–2.002989	Not switched
5	Linear	recovered	276/49	1.951670–2.001770	1.952792–2.002182	1.954844–2.002182	Not switched
6	Drop writes	lost	276/52	0.617005–2.001237	0.619248–2.002854	0.619295–2.002854	83.437–84.389
6	Log writes	recovered	260/45	0.003607–2.005085	0.004715–2.005511	1.933370–2.005511	Not switched
6	Linear	recovered	261/46	1.942407–2.013135	1.943598–2.013524	0.920646–2.013524	Not switched
7	Drop writes	lost	267/41	0.414039–2.004873	0.415025–2.005187	0.415126–2.005187	92.877–94.020
7	Log writes	recovered	281/47	0.119574–2.002861	0.120575–2.003079	0.121041–2.003079	Not switched
7	Linear	recovered	251/45	1.937550–2.005166	1.940135–2.006995	1.942750–2.160969	Not switched
8	Drop writes	lost	263/42	1.958687–2.007933	1.959705–2.008355	1.959993–2.008355	85.487–86.483
8	Log writes	recovered	262/44	1.044297–2.005459	1.045109–2.005737	1.050547–2.005737	Not switched
8	Linear	recovered	260/47	1.950299–2.002266	1.952152–2.003252	1.954262–2.003252	Not switched

Block	Arm	Outcome	W/F	W after ACK (s)	W after close (s)	F after close (s)	Switch verified after ACK (ms)
9	Drop writes	lost	263/42	1.258303–2.004289	1.259283–2.004544	1.259386–2.004544	75.889–77.013
9	Log writes	recovered	290/50	0.008957–2.468392	0.010877–2.469319	0.012066–2.183655	Not switched
9	Linear	recovered	242/42	1.942277–2.004011	1.943331–2.004432	1.944770–2.004432	Not switched
10	Drop writes	lost	296/45	1.709434–2.006333	1.710521–2.006722	1.710593–2.006722	88.547–89.584
10	Log writes	recovered	257/45	0.076972–2.007876	0.078160–2.008200	1.945861–2.008200	Not switched
10	Linear	recovered	267/49	1.919885–2.005130	1.920867–2.005356	1.723878–2.005356	Not switched

Table S5. Immediate-close mapped-device submission timing. Ranges are first lower bracket through last upper bracket, in seconds; verification is in milliseconds.

Supplementary log prefix replay methods

For each dm-log-writes trial, the original analysis recovered prefixes before the ACK marker, at that marker, and at the final recorded event. The re-analysis additionally selected the last record carrying an explicit FLUSH flag in the immutable paused-capture log and replayed that prefix onto a separate recovery derivative. FLUSH and FUA were distinguished: the final event carried FUA in all forty logs, but the last explicit FLUSH was one event earlier in thirty-nine. The kernel logs completed writes according to flush/FUA ordering [35], but these records lack completion timestamps. Prefix selection therefore refers to completed events preserved in the paused capture, not a calibrated completion time before controller dispatch. These are dependent observations within the original trials, with no new scientific units. Neither silent dropping nor prefix replay implements loss of a volatile device write cache at the fault point.

Supplementary Table S6 retains each cell separately. Every row has ten original trials, not forty additional experimental units. No formal test or extra scientific denominator is created from the dependent histories. The last literal FLUSH omits a later completed 4096-byte FUA metadata write in thirty-nine logs. One held-15-s issue overlaps the controller dispatch/quiesce bracket by approximately 0.377 ms; exact completion times relative to dispatch are absent for all logs. Recorded completion before capture is not a physical cache-loss frontier.

Condition	Original trials	Before ACK	At ACK	Last literal FLUSH	Final FLUSH/FUA	Unknown new readers
Immediate close; 2.50 s	10	0/10	0/10	10/10	10/10	0
Close at 1 s; 2.75 s	10	0/10	0/10	0/10	0/10	0
Held 5 s	10	0/10	0/10	0/10	0/10	0
Held 15 s	10	0/10	0/10	2/10	2/10	0

Table S6. Dependent log-prefix recoveries. Last literal FLUSH was separately replayed and read; final prefix is FLUSH-or-FUA-inclusive.

Object	SHA-256
Prospective protocol	a3dfb023a7fb9e4862edc4242a7cbac1b3b8865790dbcbb1156fddfd128b13b1
Frozen assignment manifest	5584eb171e36f5e3df890bc3879f2658f84414652849ef44ed6db0540f68ee42
Final scientific input lock	6b41534b91721debdfb664cf0296e22c32ac5debc22f345df75ef41914f00798
Packaged Chromium executable	839efe5fd8b6a773dd81b2e10afdc15f3c0a17316fb82b5908c0533306c4ed9e
Pinned public Ubuntu base image	1d6bffe64b848468ac97f821d369a4846d983de1800ccf6b5ec8853e85cefc55
Scientific guest worker	1179ac7e17c71781009cd5487edd96fd792394b50a8cb478c075d9c1d8acfd9b
Corrected byte-capture audit	9873f6e91abf529e0105389503769d712c9c7c6cc4bd0ff4898dca0da1935f66
Corrected timeline audit	21acbf218dce88575711863715793d9f6916ffe3b50f47ce3fd269da2c14aa1c
Append-only revalidation receipt	45630675054bbba13ab91fc3849224d5c36c44af46c0c03716c930d68a201b54

Table P1. SHA-256 identities. Protocol, manifest, and lock were rehashed for this analysis. Executable, image, worker, and audit identities are the retained acquisition or correction receipts.

Supplementary statistical details

Secondary two-sided matched-block comparisons used separate fixed Holm families: eight fault tests comparing SysRq and SIGKILL with QMP for each key; eight block-state tests comparing logging and drop-writes with linear control; twelve environment tests comparing three alternatives with custom ext4; and twelve workload

tests comparing three alternatives with quiet reference. No formal trace or byte-capture test was added. The minimum attainable unadjusted two-sided p-value with ten discordant pairs is 0.001953125. At the first Holm rank, the corresponding minima are 0.005859375 for a three-test family, 0.015625 for eight tests, and 0.0234375 for twelve. The 20-pair grid contrast can reach 0.00000190735 unadjusted; nine pairs can reach only 0.00390625. These are attainability limits, not power estimates.

Risk differences were alternative minus reference recovery proportion. Their pointwise 95% intervals used Newcombe method 10, combining the two marginal Wilson-score intervals without continuity correction [38]. This construction was not a paired or block-adjusted confidence interval; the paired randomization test and the marginal descriptive interval answer different statistical questions. We neither used its exclusion of zero as a multiplicity decision nor treated it as a simultaneous interval. The grid beyond its prespecified endpoint contrast was descriptive. Nonsignificant secondary comparisons remained in their registered families and were not treated as evidence of equivalence. A post hoc analysis ranked achieved intervals in the twenty assigned-2.0-s trials, conditional on the observed three recovery labels. All 1,140 pooled allocations and 450 allocations preserving recovery counts within close strata were enumerated. This exploratory outcome-label permutation adds exchangeability assumptions; it is not the registered blocked binary test. It neither changed a multiplicity family nor estimated a deterministic threshold. The inventory totals of recovered and lost observations were not converted into a pooled recovery rate because the chosen condition mixture determines that aggregate.

Among the twenty assigned-2.0-s trials, the three recoveries had achieved estimates of 2.003239, 2.003587, and 2.004064 s. Their pooled ranks were 8, 10, and 13, giving rank sum 31 and U = 25, near its null expectation of 25.5. The exact exploratory one-sided longer-interval p was 0.53947; the two-sided p was 1. Close-stratified enumeration also gave two-sided p = 1. Thus the recoveries did not have demonstrably longer intervals than the seventeen losses. Across the entire grid, the smallest recovered estimate was 2.003238771 s (mapping half-RTT 0.311104 ms), and the largest lost estimate was 2.009087354 s (0.409354 ms). The shorter recovered estimate remained shorter under both recorded uncertainty intervals. Three recoveries cannot establish a timing threshold.

Contrast	Recovered	Pairs	Difference [95% CI]	Exact p	Holm p
Grid: 2.4 minus 1.6 s, stratified by close time	20/20 vs 0/20	20	1.000 [0.772, 1.000]	1.9073e-06	5.722e-06
Immediate 2.50 s minus delayed-close 2.75 s	10/10 vs 0/10	10	1.000 [0.608, 1.000]	0.001953125	0.00390625
Held sweep: 15 minus 5 s	1/10 vs 0/10	10	0.100 [-0.189, 0.404]	1	1

Supplementary Table S7. Complete original primary estimates with pointwise Newcombe method 10 intervals. This marginal interval construction is not paired or block adjusted; inference uses the unchanged exact paired tests.

Supporting population / condition	R / n	Post-ACK WAL fdatasync	WAL unlink after sync
Timing: closed; 2.00 s	0 / 8	0 / 8	Not captured
Timing: closed; 2.25 s	8 / 8	8 / 8	Not captured
Close versus held: closed; 2.25 s	8 / 8	8 / 8	8 / 8
Close versus held: held; 2.25 s	0 / 8	0 / 8	0 / 8

Supplementary Table S8. Unpublished supporting syscall cells, collected 28 September 2026 before the registered campaign. Each population and condition retains its own denominator.

In the timing population, recovering closed-connection trials at 2.25 s had reconstructed second WAL fdatasync entry times of +2.007 to +2.022 s relative to the historical ACK anchor. The separate close-versus-held population had closed-connection sync entries at +2.011 to +2.029 s, followed by successful WAL unlinkat.

Times are syscall entry times for completed zero-return calls, mapped using the original paired guest realtime/monotonic sampler and the post-ACK uptime RPC. Browser ACK and uptime were separate samples; their residual latency prevents exact timer attribution. The timing population did not trace unlinkat, so its missing unlink records are not negative observations. Source trial.json placeholders, final result.json endpoints and all copied raw records are retained separately. No per-trial executable digest is present in these earlier populations; package version, launch flags, format and guest/host provenance are available. Full original VM overlays are not included in this compact supplement [26].

Supplementary byte capture methods

Retrospective target-record mapping analyzed the original 40 scientific-target_byte_diagnostics trials. The packaged binary and original raw records were not changed, and the analysis created no new scientific units, confirmatory tests or recovery endpoints. The original four cells each retain ten trials. Table S9 separates submitted history, observed later synchronization and preserved committed state; Table S10 gives exact structural locations. These observations apply to this captured family and do not map the payload-free block traces onto target sectors.

After acquisition, we decoded the existing byte-capture records without adding trials or formal tests. SQLite cell and WAL parsing followed the documented file format [28]. IndexedDB keys and Blink/V8 structured-clone values were decoded against the pinned source formats [16,29,30,31]. A match required the entire original target object, including its identifier, metadata, payload and checksum. We queried reconstructed committed views in memory and compared them with separate cell decoding and the original browser oracles. Supplementary Tables S9–S10 give the locations and condition-specific observations.

The decoder verified the exact database, object-store ID 1 (records, key path id), index ID 31 (by_trial, key path trial), decoded key, full 256-byte ASCII payload, metadata and original SHA-256 checksum. Values were uncompressed (compression_type 0), with Blink version 21 and V8 version 16. WAL validation checked headers, salts, chained checksums and commit markers. Only checksum-valid committed prefixes entered reconstructed recovery views. Frames 26–29 contain pages 5 –8, respectively: records, records_by_key, index_references and index_references_by_key; only frame 29 commits, with database size 18 pages [16,28,29,30,31].

Condition (relaxed; QMP reset)	R / n	Submitted target commit	Later WAL sync	Target DB write + DB sync	Preserved target commit
Immediate close; ACK + 2.50 s	10 / 10	10 / 10	10 / 10	10 / 10	10 / 10
Close at 1 s; ACK + 2.75 s	0 / 10	10 / 10	0 / 10	0 / 10	0 / 10
Held; ACK + 5 s	0 / 10	10 / 10	0 / 10	0 / 10	0 / 10
Held; ACK + 15 s	2 / 10	10 / 10	0 / 10	0 / 10	2 / 10

Supplementary Table S9. Retrospective byte-capture observations. Submitted commits and successful file syncs describe observed syscall histories; preserved commits describe valid reconstructed DB/WAL states, not physical persistence.

Mapped element	Verified location across all 40 submitted histories
Target SQLite record	records root page 5; row ID 2; cell index 1; cell offset 2680
Encoded key / serialized value	Page offsets 2690 (44 bytes) / 2734 (636 bytes)
Target-bearing WAL frame	Frame 26, page 5; header offset 103032; page offset 103056
Serialized value / ASCII payload in WAL	Absolute offsets 105790 / 105917
Target transaction	Frames 26–29, pages 5–8; frame 29 commits 18 database pages
Submitted transaction end	WAL offset 119512 bytes
Immediate-close database page write	4096 bytes at database offset 16384; identical target page

Supplementary Table S10. Exact target-to-page/frame mapping in submitted histories. Offsets are zero-based byte offsets; these locations do not assert that a lost trial preserved a complete target transaction.

The eligible LOST endpoint scientific-target_byte_diagnostics-00033 had a 106496-byte WAL: 25 complete valid frames followed by a 3464-byte incomplete frame-26 tail. The retained page payload contained the entire target cell, with its 44-byte key and 636-byte value verified against the contract; the neighboring sentinel cell was truncated. The tail matched the submitted frame prefix, but its complete checksum, frames 27–29 and transaction commit were unavailable. The record remained outside the last valid committed prefix. Browser and reconstructed committed-state readers both found only the sentinel. Across 28 LOST trials, 27 incomplete tails were zero-filled with no retained target-payload match and one retained the exact target value. The initial contrary expectation is preserved in TIMELINE_AND_STORAGE_INITIAL_OBSERVATIONS.json; it did not change an endpoint.

Raw streams were re-parsed and saved capture fields reproduced exactly. Raw-byte inventory hashes and historical universal-newline-normalized parser hashes were checked separately; the final SSH diagnostic contains CRLF, and neither representation was rewritten. All 640 consumed inputs were unchanged on final recheck. The decoder passed 680 checks across 40 trials, including wrong identities, truncated values, damaged WAL checksums/commit markers and rejection of uncommitted target frames. Separate SQLite 3.53.1 in-memory queries agreed with manual cell decoding and passed integrity checks. Only a derivative image's header read/write-version bytes were set to rollback mode for memory querying; originals were never modified or opened for recovery. Installed Node/V8 engines could not natively deserialize V8 version 16, so a separate native-V8 cross-check remains unavailable; no serialization header was altered to simulate that validation.

Immediate-close WAL-sync return times relative to the guest worker CLOSE_RETURNED receipt ranged from 1.999675959 to 2.009640876 s (median 2.004170126 s); database-sync returns ranged from 2.008998459 to 2.020350793 s (median 2.014486897 s). The held-15-s WAL recoveries were trials 00011 and 00030, whose main databases alone lacked the target. Target commit-write returns preceded guest ACK telemetry receipt by 3.067250–5.605042 ms, but page/guest timing brackets overlapped these returns in all 40 trials. This does not establish before-ACK ordering at the page. Pre-oracle snapshots occurred 16.744728–18.448057 s after reset dispatch, with intervening boot activity. Capture ended at reset; zero observed later syncs do not exclude unobserved operations or kernel/filesystem writeback. WAL unlink was not captured. Runtime callback attribution, fault-instant state and physical-device persistence remain unestablished.

Reproducibility files are supplied in the Byte_Capture_Reanalysis_2026-10-11 directory of the accompanying Reproducibility_Supplement.zip: ANALYSIS_PLAN.md, analyze_bytes.py, validate_decoder.py, analyze_timelines.py, RESULTS.json, TRIAL_RESULTS.csv, trials/*.json, INPUT_INTEGRITY.json, DECODER_VALIDATION.json, TIMELINE_AND_STORAGE_AUDIT.json, FINAL_VERIFICATION.json and pinned source receipts. Retained analysis-development runs are analysis executions, not scientific retries. This package is separate from public releases v1.0.0 and v1.1.0, which predate this reanalysis.

Retrospective trace-category reanalysis retained the 20 frozen assignments and their original 16 eligible endpoints, three unknown endpoints and one premeasurement slot. The raw streams contained 47,625 normally parsed trace events. A conservative prefix parser separately retained 9,184 intact event objects from malformed envelopes without repairing or overwriting raw records. These auxiliary objects did not resolve unknown outcomes. Table S11 reports native close-to-task timing in the four eligible immediate-close trials. The existing validated byte decoder also checked all 16 eligible trials' preserved DB/WAL extracts in memory. Full target contracts and committed views agreed with every original oracle. All four immediate-close recoveries contained the exact target in the main database alone; held-15-s recovery 00016 contained it only in the committed WAL-backed view. The 11 lost committed views lacked it. These extracts were obtained after reset and intervening guest activity, before the reader; they do not identify a fault-instant state or the write/sync operation producing it. FORENSIC_CROSSCHECK.json retains the exact results and input hashes.

Delayed-close recovered/ACK was 0/5, with pessimistic/optimistic missing-endpoint bounds of 0–40%; held-15-s recovered/ACK was 1/5, with bounds of 20–40%. These bounds retain the unknown outcomes; they do not replace the eligible binary denominators.

Trace ID suffix	Endpoint	Release-posted start after close (s)	First cleanup-posted start after close (s)	Cleanup evidence
00007	Recovered	2.002275	2.002479	Begin only
00010	Recovered	2.001538	2.001806	Complete, 4.860 ms
00013	Recovered	2.010446	2.010719	Begin only
00018	Recovered	2.011616	2.011885	Begin only

Supplementary Table S11. Retrospective native trace timing. Trial IDs have prefix scientific-trace_category_timelines-. Times are task start minus (IDBDatabase::close start + its recorded duration). Source labels identify posting sites, not direct callback entry. A begin-only cleanup event does not establish completion.

The four late tasks identify database_connection.cc, Release, physical line 1060, and requested delay 2000000 us. The pinned posting site binds MaybeSelfDestruct; the subsequent task begins identify backing_store_impl.cc, DestroyConnection, physical line 310. TaskAnnotator attaches pending-task posted-from metadata while wrapping its execution [16,32,33]. In trial 00010, a 4.860 ms worker task and a later 0.010 ms reply share that posting site, consistent with PostTaskAndReply; they are

not two database destructions. No database-instance ID, direct callback marker or same-trial target-write/sync link is supplied. All 16 eligible traces also contain two earlier Release-posted tasks, including held conditions, so the label alone is not a discriminator.
Fifteen eligible attempts preserve BlockingIOError write failures and malformed trace prefixes: 00001, 00002, 00005–00011, 00013, 00015–00019. All nevertheless reported no producer buffer loss. Trial 00012 has no such preserved transport error; full event completeness is not inferred. Unknown trials 00003, 00014 and 00020 also preserve write failures and partial prefixes, with no normally parsed trace events or successful trace-end receipt. Their original Required guest record missing reason remains unchanged. The controller used a 0.4 s trace-stop deadline and the worker a 5 s internal completion wait. The records do not uniquely establish the cause of each timeout. Positive retained events remain usable; event absence is not proof of absent execution.
Twelve eligible streams retain an exact trial-database open marker. Fourteen retain matching transaction ID 6 across PutOperation, DoPendingCommit and renderer onComplete. PutOperation spans 61–97 us and reports size 639; that argument is not a serialized SQLite page or the 256-byte payload size. DoPendingCommit spans 34–207 us; put start to renderer onComplete start spans 158–313 us. Fifteen streams retain process-local paired async open, add and transaction-lifetime spans. These are application/backend trace intervals, not durable-storage commit durations or substitutions for the registered ACK. No target blink.user_timing mark, direct SQLite checkpoint, WAL sync or unlink event was recorded. BlinkScheduler_PerformMicrotaskCheckpoint is unrelated to SQLite checkpointing.
Tracing.end was requested before the fault deadline and completed asynchronously. In trial 00018 the recorded late Release-posted task began about 0.81 ms after the guest end request; the request must not be treated as an instantaneous cutoff. Delayed-close coverage remains too short to observe the candidate two-second post-close task. Three immediate-close cleanup events have no recorded end, so completion and duration are unknown. The held-15-s recovered trial 00016 lacks a recorded close/cleanup signature while its telemetry confirms a held connection; no single recorded cleanup route is asserted as necessary for recovery.
Reproducibility files are supplied in the Trace_Category_Reanalysis_2026-10-11 directory of the accompanying Reproducibility_Supplement.zip: ANALYSIS_PLAN.md, reanalyse_traces.py, verify_analysis.py, RESULTS.json, TRIAL_SUMMARY.csv, TASK_TIMINGS.csv, per_trial/, event_derivatives/, INPUT_INTEGRITY.json, VALIDATION.json and pinned source receipts. All 1585 available-file manifest checks and 19 original-stream comparisons passed; 272 consumed inputs remained unchanged, and the separate validator passed 389 checks. The trace worker hash 94e2257ca12ff846470753fd71fd05ade156a311b37fdc75d25aaaf198f4b4bd and related source inputs matched FINAL_INPUT_LOCK.json. This package is separate from the earlier public releases, which predate this reanalysis.

Supplementary deviations and incidents

Incident / deviation	Effect	Disposition
Archive return-verification waiver	Historical cleanup preceded returned-byte checks; core statistical records stayed local	Disclosed; 20/41 offloaded objects later verified; remaining returns paused
Stale STATUS.md	Dashboard stopped at 420 started / 419 results	Preserved; machine ledger used: 930 slots / 929 results
Scientific worker-pin validator correction	Initial audit expected a qualification path/hash	Append-only correction to eligible loss; actual locked scientific worker and raw endpoint unchanged
Four acknowledged unknown endpoints	Three trace records missing; one SIGKILL forensic extraction failed	Excluded from binary denominators; raw evidence and endpoint bounds retained
One premeasurement trace slot	Hash-addendum validation failed before ACK; no endpoint	Slot consumed, inventoried separately, and not retried

Incident / deviation	Effect	Disposition
Last-FLUSH replay cleanup/headroom incidents	Post-read hot unplug and storage pause in dependent re-analysis	Completed prefix reads preserved; no additional scientific assignments

Supplementary Table S12. Deviations and incidents affecting the evidence record.

An archival deviation preceded analysis. The registered plan required returned-byte verification before cleanup, but later author authorization waived it after visible cloud upload and local compression checks. Core statistical records remained local. Later checks verified 20 of 41 offloaded compressed archive objects against retained SHA-256 identities and SSD readback. Further returns were paused at the author’s request, and one required chunk encountered a browser download block. The public release discloses the incomplete archive verification. The redundant local-only first-twenty archive is outside this scope; its assignments also occur in the later first-120 archive. These later checks do not erase the historical waiver.

Four acknowledged trials had unknown endpoints: three trace-family trials lacked a required guest record, and one held-15-s SIGKILL trial failed during forensic extraction. Their original records and endpoints were preserved. One additional immediate-close trace slot was consumed before measurement because its technical hash addendum failed validation; it produced no result or ACK and was not retried. The trace family's eligible counts were therefore four, three, five, and four for immediate close, delayed close, held 5 s, and held 15 s, respectively. The SIGKILL held-15-s cell had nine eligible observations and one unknown.

One delayed-close byte-capture trial initially failed a separate audit that expected the qualification worker path and hash rather than the separately locked scientific worker. The recorded guest worker matched the pre-acquisition scientific lock. An append-only validator correction changed the audit disposition to eligible loss while preserving the original verdict and raw evidence. No endpoint was fabricated, no reader was substituted, and no new trial replaced it. The corrected scientific worker-pin audit version was 1.0.0; its supporting hashes are in the appendix.

The dependent last-FLUSH re-analysis used separate replay derivatives and the pinned recovery reader. One post-read hot-unplug cleanup incident and one storage-headroom pause were preserved; completed prefix reads were not repeated. All forty raw logs and pinned reader inputs matched their original hashes after analysis. These reader operations created no target transactions, faults, or scientific assignments.

The current machine-readable ledger recorded 930 consumed slots and 929 result records. A preserved human-readable dashboard remained stale at 420 started and 419 result records. It was not used as the analysis denominator and was not silently reconciled. The machine ledger's unknown total included the premeasurement slot, while this paper distinguished four unknown result endpoints from one slot without measurement. Local re-verification completed 11,140 manifest-hash checks on the statistical core records with no mismatch; that check did not freshly hash every full disk image or verify every remote archive.

Acquisition receipt definitions and verification

The first scientific START.json timestamp is 2026-10-04T14:03:18.148755+00:00 for scientific-close_aligned_recovery_grid-00001. The final OPERATOR_INTEGRATION_RESULT.json timestamp is 2026-10-06T10:57:40.678224+00:00 for scientific-application_and_background_activity-00160. These bound the administrative acquisition interval, rather than target ACK or fault time. The public registration timestamp is 2026-10-03T18:48:13.446328 UTC. A read-only scan verified all 930 START receipts, 929 integration-completion receipts, 925 eligible endpoints and 929 observed acquisition mounts. The premeasurement trace slot had no mount observation. The administrative pre-registration REGISTRATION_RECEIPT.json remains preserved with its original pending-status fields; the later public OSF receipt supplies the registration timestamp [26].

The accompanying REPRODUCIBILITY_GUIDE.md maps the statistical, byte-capture, trace and block-prefix analyses to their scripts, dependencies and available inputs. The public statistical core is distinct from analyses requiring original capture streams or private disk volumes. Historical analysis receipts and paths remain preserved; they do not establish that every analysis can be replayed from the compact public package alone.

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